CALENDAR AND CLOCK

Odd Days, Day of the Week, Hands and Directions

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 20

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: a calendar question (English 2022 Q36) and a clock-direction question (English 2019 Q57) appear.

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Two questions appear (English 2022 Q36 on calendar and English 2019 Q57 on clock). Both are solved below. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 2 PYQs and the rules they used
2	Foundation	Odd-day method, leap years, clock angles and directions
3	Solved PYQs	2 PYQs with step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

One calendar and one clock question

Paper	Q.	Question	Rule used	Ans
English 2022	36	1 February 1948 was a Sunday; what day was 1 February 1952?	Leap-day count, odd days	C (Friday)
English 2019	57	Clock shows 3:00; minute hand points North; where does the hour hand point?	Clock directions	A (East)
Mathematics 2024 / Science 2024 / Social Studies 2023	–	none		

PYQ INSIGHT: Both are short **one-step** questions. The calendar question needs only the count of leap days between the two dates; the clock question needs only the position of the numbers on the dial. Expect one of these in a paper, often with a leap year hidden in the gap.

Pattern	PYQs	Priority
Day of the week after some years (leap-day count)	1	★★★ Must
Direction of clock hands	1	★★★ Must
Day of the week for a given date (odd days)	0	★★ Expected
Angle between hands, mirror time of a clock	0	★★ Expected

CHAPTER 2

Foundation

Odd days and the dial

2.1 Days and odd days

Odd days are the days left over after removing complete weeks. Number the days Sunday = 0, Monday = 1, ... Saturday = 6.

Period	Odd days
Ordinary year (365 days)	1
Leap year (366 days)	2
100 years	5
200 years	3
300 years	1
400 years	0

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Odd days	3	0 (1 if leap)	3	2	3	2	3	3	2	3	2	3

NOTE: A year is a leap year if it is divisible by 4. A century year is a leap year only if it is divisible by 400 (1900 was not a leap year; 2000 was).

2.2 Day after some years (the PYQ type)

Step	Do this
1	Count the years between the two dates.
2	Count how many **29 February** days fall between the two dates (a leap day counts only if it lies between them).
3	Total shift in days = years + leap days. Divide by 7; the remainder moves the day forward.
4	Check the end points: a date before 29 February in a leap year does not include that leap day.

SHORTCUT: 1 Feb 1948 to 1 Feb 1952 is 4 years with one leap day (29 Feb 1948), so the shift is 4 + 1 = 5 days: Sunday + 5 = Friday (English 2022 Q36). The 29 Feb 1952 is after 1 February 1952, so it is not counted.

2.3 Clock

Rule	Formula / fact
Angle between hands at H:M	$\|30H - 5.5M\|$ (take 360° minus it if it is more than 180°)
Hands coincide	11 times in 12 hours
Hands at right angles	22 times in 24 hours
Hands in a straight line	22 times in 24 hours
Mirror image of a clock	Mirror time = 11:60 − actual time (e.g. 4:25 becomes 7:35)
Minute hand moves	6° per minute; hour hand 0.5° per minute

2.4 Directions on a clock

Treat the dial like a compass in the position stated: the number the minute hand points to fixes North (usually 12). Then 3 is East, 6 is South and 9 is West. Each hour mark is 30°; going clockwise from North, 1 is North-North-East (30°), 3 is East (90°).

If 12 points to	3 points to	6 points to	9 points to
North	East	South	West

SHORTCUT: At 3:00 the minute hand is on 12 and the hour hand is on 3, a quarter turn clockwise. If the minute hand points North, the hour hand points **East** (English 2019 Q57). At 9:00 it would point West.

CHAPTER 3

Solved PYQs

Both questions with step-by-step reasoning

English 2022 Q36

In the year 1948, the first day of February was Sunday. What was the day on 1 February in 1952? (A) Saturday (B) Monday (C) Friday (D) Sunday

1. From 1 February 1948 to 1 February 1952 is 4 years.
2. The only 29 February between the two dates is 29 February 1948 (it comes after 1 February 1948). The 29 February 1952 comes after the second date, so it is not counted.
3. Shift = 4 years + 1 leap day = 5 days.
4. Sunday + 5 days = Friday.
Answer: Option C (Friday)

English 2019 Q57

The clock in hand is showing 3:00 am. If the minute hand points towards North, then in which direction does the hour hand point? (A) East (B) North (C) South (D) West

1. At 3:00 the minute hand is on 12 and the hour hand is on 3.
2. The minute hand points North, so the 12 marks North.
3. Going clockwise, 12 is North, 3 is East. So the hour hand points to the East.
Answer: Option A (East)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Leap-day count:** only 29 Februaries that lie between the two dates count.
- **Odd-day arithmetic:** work with the remainder after dividing by 7.
- **Same day:** if the shift is 0 mod 7, the day is unchanged.
- **Mirror clock:** subtract the time from 11:60.
- **Directions:** treat the number the minute hand points to as North, then count clockwise in quarter turns.
- **Time:** 45 seconds per question.

Trap	How to avoid it
Counting a leap day that is outside the dates	A 29 February counts only if it lies between the two dates.
Treating 1900 as a leap year	Century years need divisibility by 400.
Moving the day backwards	For an earlier date, subtract the shift.
Reading clock directions anticlockwise	Numbers run clockwise: 12, 3, 6, 9.
Angle above 180°	Use 360° minus it.

Scope

Related PYQs / topics	Where
Direction and distance questions (walks and turns)	Direction (earlier topic)
Ranking questions	Ranks (earlier topic)

CHAPTER 5

Practice Set

6 original questions; answers on the next page

A. Calendar

Q1. What was the day of the week on 15 August 1947?

A. Saturday B. Sunday C. Friday D. Tuesday

Q2. What was the day of the week on 1 January 2001?

A. Monday B. Tuesday C. Wednesday D. Friday

Q3. 1 March 2016 was a Tuesday. What day of the week was 1 March 2019?

A. Saturday B. Sunday C. Tuesday D. Friday

Q4. 15 August 2018 was a Wednesday. What day of the week was 15 August 2020?

A. Sunday B. Saturday C. Monday D. Wednesday

B. Clock

Q5. What is the angle between the hour hand and the minute hand at 4:20?

A. 10° B. 20° C. 5° D. 15°

Q6. The clock shows 9:00 and the hour hand points towards South. In which direction does the minute hand point?

A. North B. East C. West D. South-west

Answer key

Q	Answer	Reason
1	C	1946 complete years: 1600 years = 0 odd days, 300 years = 1, 46 years with 11 leap years = 57 days = 1, total 2. Days from 1 January to 15 August 1947 = 227 = 3 odd days. 2 + 3 = 5, which is Friday.
2	A	2000 complete years have 0 odd days and 1 January is the first day of the year, so the day is 0 + 1 = 1, which is Monday.
3	D	29 February 2016 is before 1 March 2016, so no leap day lies between 1 March 2016 and 1 March 2019. Shift = 3 years = 3 days: Tuesday + 3 = Friday.
4	B	29 February 2020 lies between the two dates, so the shift is 2 years + 1 leap day = 3 days: Wednesday + 3 = Saturday.
5	A	$\|30 \times 4 - 5.5 \times 20\| = \|120 - 110\| = 10°$.
6	C	At 9:00 the minute hand is on 12 and the hour hand is on 9. From 9 clockwise to 12 is a quarter turn. A quarter turn clockwise from South is West.

CHAPTER 6

Quick Revision

One page for the last day

Item	Rule
Odd days	Ordinary year 1, leap year 2, 100 years 5, 200 years 3, 300 years 1, 400 years 0
Day after n years	Shift = n + leap days between the dates, mod 7
Leap year	Divisible by 4; century only if divisible by 400
Angle between hands	$\|30H - 5.5M\|$
Mirror clock	11:60 − time
Clock directions	Minute-hand number = North; 3 East, 6 South, 9 West

PYQ RULE: Both PYQs are one-step: count leap days between the dates (Eng 2022 Q36) or read the dial like a compass (Eng 2019 Q57).

ROUTINE: Exam-day routine: count years and leap days; reduce mod 7; for clocks, fix North by the minute hand and count quarter turns. 45 seconds per question.